

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0308	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – ConstraintBased Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	2024	Date:	

Code of Conduct

I B V Chaitanya Reddy-192425376 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _chaitanya

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. Reference Resolution and Discourse Coherence Given

the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving." Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility • Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation Conversation

History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review ○ improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.
5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water." Note:

The word "bank" can refer to:

- A financial institution
 - The side of a river
- Constraints:**

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z) move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts

Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided
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Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	3	3	3	4	3	16	80%
2	3	3	3	3	3	15	
3	3	3	3	3	4	16	

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____

Answers:

Q1. Reference Resolution and Discourse Coherence Given

paragraph:

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Identify all the referring expressions and their possible antecedents

A referring expression is a word or phrase that refers to an entity already mentioned or understood from the discourse.

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	A book on Artificial Intelligence
they	Ravi and Arun

the book	A book on Artificial Intelligence
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The pronouns “He” and “him” initially have more than one possible antecedent because both Ravi and Arun are singular male persons.

b) Apply the given constraints to resolve the references

The following constraints are used for reference resolution:

1. Gender and number agreement
 2. Recency
 3. Grammatical role
 4. Semantic compatibility
 5. Discourse coherence
- Reference resolution for “He” Sentence:**

“He was looking for a book on Artificial Intelligence.” The possible antecedents are Ravi and Arun.

Ravi is male and singular, so he agrees with “He”. Arun is also male and singular, so he also satisfies the agreement constraint.

However, Ravi is the subject of the preceding sentence and is the main discourse participant being continued. Therefore, based on grammatical role and discourse coherence, Ravi is the preferred antecedent. Thus:

He → Ravi

Reference resolution for “him” Sentence:

“Arun helped him find it.”

The possible antecedents are Ravi and Arun.

Ravi is male and singular and is semantically compatible with the action of being helped by Arun. Arun is also male and singular, but “Arun helped Arun” is not the intended interpretation in this context.

Therefore: him

→ Ravi

Reference resolution for “it”

The previous sentence introduces “a book on Artificial Intelligence”.

The verb “find” is semantically compatible with a physical object such as a book. Therefore:

it → book on Artificial Intelligence

Reference resolution for “they” The

sentence is:

“Later, they discussed the book before leaving.”

The plural pronoun “they” refers to the two people involved in the previous events.

Therefore:

they → Ravi and Arun

Reference resolution for “the book”

“The book” refers to the previously introduced book on Artificial Intelligence.

Therefore: the book → book on Artificial
Intelligence

c) Constraint analysis table

Referring expression	Possible antecedent	Validity	Reason
He	Ravi	Valid	Gender and number agree; Ravi is the previous subject and maintains discourse continuity
He	Arun	Possible but less preferred	Gender and number agree, but Ravi has stronger grammatical and discourse preference
Him	Ravi	Valid	Gender and number agree and the reference is semantically appropriate
Him	Arun	Less preferred	Arun helping himself is not the intended semantic interpretation
It	Book on artificial intelligence	Valid	The book was previously introduced and is compatible with the verb “find”

f) Discuss the ambiguity if “He was looking for a book” appeared before introducing Arun

If the sentence “He was looking for a book” appeared immediately after introducing Ravi and before Arun was introduced, “He” would most naturally refer to Ravi because Ravi would be the only previously introduced suitable male antecedent.

For example:

“Ravi went to the library. He was looking for a book. Later, Arun arrived.”

In this case, the reference is clear because Ravi is the only suitable antecedent.

However, if the pronoun “He” appeared without any previously introduced male entity, the reference would be unresolved. If multiple suitable male entities had already been introduced, the pronoun could become ambiguous.

Therefore, reference resolution depends on the availability of antecedents and the surrounding discourse context.

Q2. Dialog Act Recognition and Response Generation Conversation

history:

Student: “I submitted my assignment, but I am worried that I may have made many mistakes.”

a) Identify the dialog acts required The required dialog acts are:

1. Reassure
2. Advise

The reassurance dialog act should address the student's worry and provide emotional support. The advice dialog act should suggest a constructive action, such as reviewing the assignment and using feedback to improve.

Therefore:

Student statement



Reassure + Advise



Positive and constructive response

b) Identify the important entities that must be maintained

Entity	Role in the conversation
Student / You	Person receiving the response
Assignment	Work submitted by the student
Mistakes	Possible errors in the assignment
Review	Action recommended to identify errors

Feedback	Information that can help identify areas for improvement
Improve	Desired outcome of using feedback

The response should continue discussing these entities rather than introducing unrelated topics.

c) Generate three possible responses satisfying all constraints

Response 1

It is understandable to feel worried, but you can become more confident by reviewing your assignment carefully. Use the feedback you receive to identify mistakes and improve your future work.

Response 2

Do not worry too much about possible mistakes in your assignment, as mistakes are a normal part of learning. Review the assignment and use the feedback to improve your work and become more confident.

Response 3

You have already completed an important step by submitting your assignment, so there is no need to worry excessively. Review the mistakes identified through feedback and use them as an opportunity to improve.

d) Compare the responses

Criterion	Response 1	Response 2	Response 3
Coherence	Excellent	Excellent	Excellent
Constraint satisfaction	Excellent	Excellent	Excellent
Politeness	Excellent	Excellent	Excellent
Relevance	Excellent	Excellent	Excellent
Entity consistency	Maintained	Maintained	Maintained
Positive tone	Maintained	Maintained	Maintained
Required keywords	Satisfied	Satisfied	Satisfied

Response 1

Response 1 directly reassures the student and then provides advice. It maintains the entities assignment, mistakes, review, feedback, and improvement. The response also has a clear cause-and-effect relationship because reviewing the assignment and using feedback can help improve future work.

Response 2

Response 2 provides strong reassurance by explaining that mistakes are a normal part of learning. It then gives practical advice to review the assignment and use feedback for improvement. The response is polite and relevant.

Response 3

Response 3 reassures the student by emphasizing that submitting the assignment is already a completed task. It then recommends using feedback and identified mistakes as opportunities for improvement.

e) Select the best response and justify the answer Response 1 is selected as the best response.

The reasons are:

1. It directly addresses the student's worry.
2. It provides reassurance without ignoring the student's concern.
3. It gives a practical recommendation to review the assignment.
4. It maintains entity coherence using assignment, mistakes, feedback, and improvement.
5. It contains the required keywords review, improve, confident, and feedback.
6. It maintains a positive and polite tone.
7. It does not provide contradictory advice.
8. It contains two sentences, satisfying the required response length.
9. It establishes a clear relationship between reviewing the assignment, receiving feedback, and improving future work.

Therefore, Response 1 provides the most balanced combination of reassurance, advice, coherence, relevance, and constraint satisfaction.

f) Explain what happens if entity coherence is not maintained

Entity coherence means maintaining consistent references to the same entities throughout a conversation or discourse.

If entity coherence is not maintained, the response can become confusing, ambiguous, irrelevant, or semantically inconsistent.

For example:

Student: "I submitted my assignment, but I am worried about mistakes." Unrelated response:

"You should repair the computer before submitting it."

In this response, the discussion moves from the assignment to a computer without any connection. The pronoun "it" also becomes unclear.

Therefore, maintaining entity coherence is essential for generating responses that are meaningful, relevant, and easy to understand.

Q3. Word Sense Disambiguation and Meaning Representation Source

sentence:

“Priya sat on the bank and watched the boats moving across the water.”

a) Identify the possible meanings of “bank”

The word “bank” is ambiguous because it can have different meanings depending on its context.

The two possible meanings are:

1. Financial institution

A financial organization where people can deposit money, withdraw money, maintain accounts, and perform financial transactions.

2. Side of a river

The land beside a river or other body of water.

Therefore:

BANK

/ \

/ \

Financial institution River bank

b) Select the correct sense and justify the decision using contextual constraints

The correct sense of “bank” is the side of a river.

The surrounding words provide strong contextual evidence:

- “boats”
- “water”
- “moving across”
- “sat”
- “watched”

These words describe a waterside environment. A person can naturally sit on the side of a river and watch boats moving across the water.

The financial meaning is not compatible with the context because there are no words related to money, accounts, deposits, withdrawals, or financial transactions.

Constraint analysis

Constraint	Observation	Result
Contextual information	Boats and water are mentioned	River bank

Semantic compatibility	Sitting beside water is natural	River bank
Action compatibility	Boats can be watched from a river bank	River bank
Discourse coherence	All elements describe a waterside scene	River bank
Financial context	No financial activities are mentioned	Financial sense rejected

Therefore:

bank → side of a river

The correct sense is selected through contextual and semantic constraints.

c) Predicate logic representation

The sentence can be represented using suitable predicates.

Let:

- Priya be represented by p
 - Bank be represented by b
 - Boat be represented by z
 - Water be represented by w
- The predicates are:

Person(Priya)

Bank(b)

Water(w)

sit(Priya)

beside(Priya,b)

boat(z) move(z)

across(z,w)

watch(Priya,z)

The combined predicate logic representation is:

Person(Priya) $\wedge$ Bank(b) $\wedge$ Water(w)

$\wedge$ sit(Priya) $\wedge$

beside(Priya,b) $\wedge$ boat(z)

$\wedge$ move(z) $\wedge$

across(z,w) $\wedge$

watch(Priya,z)

This representation captures the important entities, actions, and semantic relationships in the sentence.

The representation indicates that Priya is sitting beside a river bank and watching boats that are moving across the water.

d) Generate a simple English paraphrase without ambiguity

Priya sat on the side of a river and watched boats moving across the water.

This paraphrase removes the ambiguity of “bank” by explicitly stating “the side of a river”.

e) Explain how Word Sense Disambiguation improves Machine Translation

Word Sense Disambiguation is the process of identifying the intended meaning of an ambiguous word using its context.

In the given sentence, “bank” has two possible meanings: bank

└── Financial institution

└── Side of a river

The words “boats” and “water” provide contextual evidence that the intended meaning is the side of a river.

Without WSD, a machine translation system may select the financial meaning of “bank”. This could produce an incorrect translation because the selected meaning would not match the surrounding context.

With WSD, the system first identifies the correct meaning and then translates the sentence accordingly.

WSD improves machine translation in the following ways:

1. It reduces lexical ambiguity.
2. It preserves the intended meaning of the source sentence.
3. It improves semantic accuracy.
4. It helps select the appropriate target-language word.
5. It produces more natural translations.
6. It maintains semantic relationships between words.
7. It reduces incorrect translations caused by multiple word meanings.

For example:

Source:

Priya sat on the bank and watched the boats.

↓

WSD:

bank = side of a river

↓

Translation:

Use the target-language word meaning river bank rather than the word meaning financial institution.

Therefore, WSD is an important component of semantic processing in machine translation because it ensures that the intended meaning of an ambiguous word is preserved during translation.